

1. What each Magbot is in the code

Each Magbot is treated as a **rigid circular disk**.

It has:

```
Nbots = 100
Nmag = 6
R_bot = 1.0
R_ring = 0.78
```

Meaning:

- Nbots = 100: total 100 Magbots.
- Nmag = 6: each Magbot has 6 magnetic sites.
- R_bot: radius of the Magbot body.
- R_ring: radius where magnets are placed around the center.

This matches the paper idea where Magbots have up to **six magnetic rods around their perimeter**.

2. Position and orientation of each Magbot

Each Magbot has:

```
pos[i] = [x_i, y_i]
theta[i] = body angle
phi[i, k] = internal magnet rotation angle
```

So for bot *i*:

- pos[i] tells where the bot center is.
- theta[i] tells the body orientation.
- phi[i, k] tells how the kth internal magnet has rotated relative to the body.

The magnet angle is:

```
ang = theta[:, None] + ALPHA[None, :] + phi
```

Meaning:

$magnet\ angle = \theta_i + \alpha_k + \phi_{ik}$ | $text\{magnet\ angle\} = \theta_i + \alpha_k + \phi_{ik}$ magnet
angle = $\theta_i + \alpha_k + \phi_{ik}$

where:

θ_i | θ_i : body orientation,
 α_k | α_k : fixed angular position of kth magnet,
 ϕ_{ik} | ϕ_{ik} : extra internal magnet phase.

3. Chiral rotation physics

The paper's Magbots rotate because vibration motors and tilted brushes convert vibration into clockwise or counterclockwise rotation.

In code:

```
omega_intrinsic = chirality * (omega_base + omega_light_gain * I)
```

Formula:

$\omega_i = \chi_i(\omega_0 + \beta I_i)$ | $\omega_i = \chi_i(\omega_0 + \beta I_i)$ $\omega_i = \chi_i(\omega_0 + \beta I_i)$

where:

ω_i | ω_i : angular velocity of Magbot i ,
 $\chi_i = -1$ | $\chi_i = -1$: clockwise,
 $\chi_i = +1$ | $\chi_i = +1$: counterclockwise,
 ω_0 | ω_0 : base rotation speed,
 I_i | I_i : local light intensity,
 β | β : light sensitivity.

Then body angle updates as:

```
theta = theta + (omega_intrinsic + mu_r * T_body) * dt
```

Formula:

$$\theta_i(t+\Delta t) = \theta_i(t) + (\omega_i + \mu_r \tau_i) \Delta t + \text{rotational noise}$$

$$\theta_i(t+\Delta t) = \theta_i(t) + (\omega_i + \mu_r \tau_i) \Delta t + \text{rotational noise}$$

This means the bot rotates because of:

- its own motor/chirality,
- torque from magnetic interactions,
- random noise.

4. Light-field physics

The yellow circle is the **light field**. In the paper, the LED platform controls Magbot motion through photoresistors.

In code, light is modeled as a Gaussian spot:

$$I(x) = I_0 + A \exp\left(-\frac{(x - x_L)^2}{2\sigma^2}\right)$$

$$I(\mathbf{x}) = I_0 + A \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_L\|^2}{2\sigma^2}\right)$$

where:

- I_0 : background light,
- A : light strength,
- x_L : light spot center,
- σ : light spot width.

The light spot moves in a circle:

$$x = \text{radius} * \cos(\text{angle})$$

$$y = \text{radius} * \sin(\text{angle})$$

Formula:

$$x_L(t) = R \cos(2\pi t/T)$$

$$y_L(t) = R \sin(2\pi t/T)$$

So the yellow light circle keeps moving around.

5. Translational motion physics

Earlier the Magbots were only rotating. So I added stronger translational motion.

The position update is:

```
pos = pos + (mu_t * F + F_light + v0[:, None] * n) * dt
```

Formula:

$$\mathbf{x}_i(t+\Delta t) = \mathbf{x}_i(t) + [\mu_t \mathbf{F}_i + \mathbf{F}_{\text{light},i} + v_{i0} \mathbf{n}_i] \Delta t + \text{noise}$$

$$\mathbf{x}_i(t+\Delta t) = \mathbf{x}_i(t) + [\mu_t \mathbf{F}_i + \mathbf{F}_{\text{light},i} + v_{i0} \mathbf{n}_i] \Delta t + \text{noise}$$

where:

$\mathbf{F}_i$: magnetic + repulsive force,
 $\mathbf{F}_{\text{light},i}$: drift toward bright light,
 v_{i0} : self-propulsion,
 $\mathbf{n}_i = (\cos \theta_i, \sin \theta_i)$: heading direction.

The speed is:

```
v0 = v_base + v_light_gain * I
```

Formula:

$$v_i = v_{\text{base}} + \gamma I_i$$

So bots move faster in brighter light.

6. Light-gradient drift

To make bots move toward the light, I added:

```
F_light = light_drift_strength * gradI
```

Formula:

$$F_{light} = k_L \nabla I \quad \mathbf{F}_{light} = k_L \nabla I$$

For Gaussian light:

$$\nabla I = \frac{I}{\sigma^2} (x - x_0) \quad \nabla I = \frac{I}{\sigma^2} (x - x_0)$$

So if a bot is near the light spot, it experiences a drift toward the brighter region.

This is a simplified control rule, not exact experimental physics, but it represents the paper's idea of light-guided control.

7. Magnetic interaction physics

Each Magbot has 6 magnetic sites. The code checks magnet pairs between nearby bots.

The simplified magnetic energy used is:

$$U = -k_m e^{-r^2/r_0^2} \cos(\Delta\phi) \quad U = -k_m e^{-r^2/r_0^2} \cos(\Delta\phi)$$

where:

k_m : magnetic strength,

r : distance between two magnetic sites,

r_0 : interaction length scale,

$\Delta\phi$: difference between magnet orientations.

Interpretation:

If $\cos(\Delta\phi) > 0$, magnets are favorably aligned and attract.

If $\cos(\Delta\phi) < 0$, they repel or interact unfavorably.

Interaction becomes weaker with distance because of e^{-r^2/r_0^2} .

The force is derived from the energy:

$$\mathbf{F} = -\nabla U \quad \mathbf{F} = -\nabla U$$

In code:

```
coeff = (2.0 * k_m / r0**2) * weight * cos(delta)
f_pair_on_i = coeff * r
```

This creates attraction when magnet phases are aligned.

8. Internal magnet torque

The internal magnet phase torque is:

```
tau = k_m * weight * sin(delta)
```

Formula:

$$\tau_{\phi} = k_m e^{-r^2/r_0^2} \sin(\Delta\phi) \quad \tau_{\phi} = k_m e^{-r^2/r_0^2} \sin(\Delta\phi)$$

This makes internal magnets rotate to reduce magnetic energy.

So magnets are not fixed permanently; they can adjust their orientation.

9. Body torque

When a magnetic force acts away from the center of a Magbot, it creates torque.

Formula:

$$\tau_{body} = r_{lever} \times F \quad \tau_{body} = \mathbf{r}_{lever} \times \mathbf{F}$$

In code:

```
T_body[i] += cross2d(lever, force)
```

This torque rotates the entire Magbot body.

So magnetic interaction affects:

translation,
body rotation,
internal magnet rotation.

That is why the simulation is physically richer than simple particle attraction.

10. Hard-core repulsion

Bots should not overlap. So if two disks overlap, the code applies repulsion:

```
overlap = 2*R_bot - d
f_i = -k_rep * overlap * u
```

Formula:

$$F_{rep} = -k_{rep}(2R-d)r^{\wedge} \hat{\mathbf{r}} \quad \mathbf{F}_{rep} = -k_{rep}(2R-d)\hat{\mathbf{r}}$$

This pushes overlapping bots apart.

Then `resolve_overlaps()` also directly separates disks if they still overlap.

11. Wall reflection

If a bot hits the boundary, it reflects.

For x-wall collision:

$$\theta \rightarrow \pi - \theta \quad \theta \rightarrow \pi - \theta$$

For y-wall collision:

$$\theta \rightarrow -\theta \quad \theta \rightarrow -\theta$$

This is like elastic reflection from walls.

12. Why cell-list is used

Checking all pairs for 100 bots is possible, but slow for larger systems.

All-pair checking scales as:

$$O(N^2)$$

Cell-list reduces this because each bot only checks nearby bots.

So the code divides space into cells and checks neighboring cells only. Your original code also used this efficient idea.

13. What green interaction lines mean

The green lines are display-only lines. They show magnet pairs that are close enough to interact.

They are drawn when:

```
distance < r_cut  
strength > threshold
```

Strength is approximately:

$$S = e^{-r^2/r_0^2} |\cos(\Delta\phi)|$$

So darker/thicker green lines mean stronger magnetic interaction.

14. What this simulation represents

This code is not an exact replica of the paper. It is a **simplified physics-inspired model**.

It captures:

- many Magbots,
- chiral rotation,

light-controlled activity,
magnetic attraction/torque,
internal magnet rotation,
active motion,
collisions,
interaction visualization.

The paper's real system additionally has:

real vibration motors,
real tilted brushes,
real magnetic rods,
real LED feedback platform,
camera tracking,
physical self-assembly into phases like active crystal, liquid-like, gas-like, etc.

Best report line:

This simulation models Robo-Matter as a 2D active magnetic disk system. Each Magbot is represented as a rigid circular body with six internal magnetic sites. The bots undergo chiral rotation, light-controlled activation, translational active motion, magnetic force, body torque, and internal magnet phase torque. A Gaussian moving light field imitates programmable LED control, while cell-list neighbor search makes the simulation scalable to 100 bots.

Core Physical Competition

Activation / Agitation vs *Magnetic Binding / Ordering* | *text{Activation / Agitation}* | *quad* vs | *quad* | *text{Magnetic Binding / Ordering}* | *Activation / Agitation* vs *Magnetic Binding / Ordering*

If motion dominates → bots separate.

If magnetic attraction dominates → bots cluster/order.

This is exactly why different phases appear.

1. What Each Parameter Does

A. Motion / Activity Parameters

These make Magbots move and rearrange.

v_base

v_base

Base translational speed.

Higher v_base means bots travel faster.

- low → bots stay near neighbors
- high → bots break clusters and roam

v_light_gain

v_light_gain

Extra speed due to light.

$$v = v_{base} + v_{light} \quad | \quad v = v_{\{base\}} + v_{\{light\}} \quad | \quad v = v_{base} + v_{light} \quad |$$

Higher means bots speed up strongly in bright light.

omega_base

omega_base

Base spinning speed (CW or CCW).

Higher means bots rotate faster.

Effects:

- frequent direction changes
- more agitation
- less stable binding

omega_light_gain

Extra spin due to light.

$$\omega = \omega_{base} + \omega_{light} \quad | \quad \omega = \omega_{\{base\}} + \omega_{\{light\}} \quad | \quad \omega = \omega_{base} + \omega_{light} \quad |$$

Bots near light rotate faster.

B. Magnetic Attraction Parameters

These make bots bind.

k_m

Magnetic strength.

Most important clustering parameter.

- low → weak interactions
- high → strong clusters / jammed groups

r0

Interaction decay scale.

In your force:

$$e^{-r^2/r_0^2} e^{-r^2/r_0^2} e^{-r^2/r_0^2}$$

Larger r0 means force decays slower.

So magnets feel each other farther away.

r_cut

Maximum cutoff distance.

If farther than this, no interaction.

Large r_cut = more interacting neighbors.

C. Density / Crowding Parameters

box

System size.

Bots move in square:

[-box, box]

Area:

$$A = (2\text{box})^2 A = (2\text{box})^2 A = (2\text{box})^2$$

Small box = crowded.

Large box = spacious.

Crowding increases collisions and clustering.

Nbots

Number of bots.

More bots = higher density if box fixed.

D. Noise Parameters

Dt

Random translational shaking.

Dr

Random body rotation.

Dphi

Random magnet rotation.

Higher noise destroys order.

E. Repulsion Parameters

k_{rep}

Prevents overlap.

Too low = bots penetrate.

Too high = hard collisions.

2. Useful Dimensionless Idea

Think:

$$\text{Phase} \sim \frac{\text{Binding Energy}}{\text{Activity}} \quad \text{Phase} \sim \frac{\text{Binding Energy}}{\text{Activity}}$$

Where:

$$\text{Binding} \sim k_m$$

$$\text{Activity} \sim v_{base} + \omega_{base}$$

If:

$$k_m \gg v, \omega \quad k_m \gg v, \omega$$

solid-like / glass / crystal

If:

$$k_m \approx v, \omega \quad k_m \approx v, \omega$$

liquid

If:

$$k_m \ll v, \omega \quad k_m \ll v, \omega$$

gas

3. Conditions for 4 Phases

Phase 1: Active Glass-like

Condition:

Strong attraction + low motion + crowded + disorder trapped.

$$k_m \text{ high}, v, \omega \text{ low} \quad k_m \text{ high}, v, \omega \text{ low}$$

Bots stick in irregular jammed clusters.

No easy rearrangement.

Use:

k_m high
v_base low
omega_base low
box medium-small

Phase 2: Active Crystal Phase

Condition:

Moderate motion + strong enough attraction + enough time to reorder.

$k_m \text{ high}, v, \omega \text{ moderate}$
 $k_m \text{ high}, v, \omega \text{ moderate}$

Bots form ordered lattice / honeycomb / symmetric packing.

Need motion to anneal defects.

Too low motion gives glass.

Too high motion melts crystal.

Use:

$k_m \text{ high}$
 $v \text{ moderate}$
 $\omega \text{ moderate}$
 box moderate
 low noise

Phase 3: Liquid-like Phase

Condition:

Activity comparable to attraction.

$k_m \approx v, \omega$
 $k_m \approx v, \omega$

Clusters form and break continuously.

Bots flow like fluid.

Use:

k_m medium
v medium-high
omega medium

Phase 4: Gas-like Phase

Condition:

Very high motion + weak attraction + spacious system.

$$v, \omega \gg kmv, |\omega| \gg k_m v, \omega \gg km$$

Bots mostly independent.

Random roaming.

Use:

v high
omega high
k_m low
box large

4. Important Transition Logic

Glass → Crystal

Increase motion slightly.

Why?

Bots escape local traps and reorder.

Crystal → Liquid

Increase motion more.

Bonds break/reform.

Liquid → Gas

Increase motion further or reduce k_m .

No sustained clusters.